UHPC Compressive Strength Prediction

A Machine Learning Approach

AI-generated technical companion to the project notebooks. Provides an expanded narrative of the methodology, decisions, and results documented in the analysis.

Technical Analysis Report

Exploratory Analysis • Model Development • Interpretation

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January 2026

Table of Contents

Executive Summary
1. Introduction
1.1 Domain Background
1.2 Project Scope and Objectives
1.3 Pipeline Design Philosophy
2. Technical Decisions and Methodology Rationale
2.1 Why RMSE as the Primary Metric
2.2 Why Five-Fold Cross-Validation
2.3 Why StandardScaler Inside a Pipeline
2.4 Why Flag-and-Filter for Data Cleaning
2.5 Why These Five Models
3. Phase 1: Exploratory Data Analysis
3.1 Dataset Overview and Data Quality
3.2 Target Variable Analysis
3.3 Feature Distributions and Characteristics
3.4 Feature-Target Relationships
3.5 Multicollinearity Assessment
4. Phase 2: Model Development
4.1 Data Preparation
4.2 Baseline Model Comparison
4.3 Hyperparameter Tuning
4.4 Final Model Evaluation
5. Phase 3: Model Interpretation (SHAP Analysis)
5.1 SHAP Methodology
5.2 Global Feature Effects
5.3 Individual Prediction Analysis
6. Practical Implications
6.1 Who Would Use This Model
6.2 What 5.93 MPa Error Means in Practice
6.3 Actionable Insights from the Analysis
7. Conclusions, Limitations, and Future Work
7.1 Summary of Key Findings
7.2 Limitations
7.3 Future Work
References

Executive Summary

This report documents a complete machine learning pipeline for predicting the compressive strength of Ultra-High-Performance Concrete (UHPC) from mix design parameters. The project spanned three analysis phases—exploratory data analysis, model development, and model interpretation—each captured in a dedicated Jupyter notebook with explicit data handoffs between phases to ensure full reproducibility. The work was motivated by a practical question: given a set of concrete mix ingredients and curing conditions, can we reliably predict how strong the resulting concrete will be?

The analysis used a publicly available dataset of 810 UHPC mix designs (Kashem et al., 2023), each described by 13 features including cement content, water, aggregate, silica fume, fiber reinforcement, curing temperature, and age at testing. After data quality validation removed 18 duplicate records, 792 observations remained for analysis. The target variable—compressive strength measured in megapascals (MPa)—ranged from 29 to 221 MPa, with a mean of 122 MPa. A 150 MPa threshold served as the benchmark for UHPC-grade classification throughout the project.

Five regression models were evaluated: Linear Regression, Ridge, LASSO, Random Forest, and XGBoost. Tree-based models dramatically outperformed linear approaches, reducing prediction error by roughly threefold—a result that confirmed the presence of strong non-linear relationships between mix components and strength. After hyperparameter tuning via GridSearchCV, the final XGBoost model achieved an RMSE of 5.93 MPa on a held-out test set—approximately 4.9% of the mean compressive strength—with an R^2 of 0.978. SHAP (SHapley Additive exPlanations) analysis revealed that curing age, silica fume content, and fiber reinforcement were the most influential predictors, with age alone capable of shifting predictions by over 40 MPa in either direction.

The results demonstrate that gradient-boosted tree models can capture the complex, non-linear interactions inherent in concrete mix design with high accuracy. The analysis also identified several limitations: the dataset is moderately sized at 792 records, fiber and aggregate types are unspecified, and bootstrap prediction intervals showed lower-than-nominal coverage (81.8% vs. 95%), suggesting that uncertainty quantification could be improved with conformal prediction methods.

1. Introduction

1.1 Domain Background

Ultra-High-Performance Concrete (UHPC) represents the frontier of cementitious materials, achieving compressive strengths that far exceed those of conventional concrete. The Federal Highway Administration defines UHPC as concrete achieving at least 150 MPa in compressive strength—roughly five to ten times stronger than standard structural concrete, which typically falls in the 20–40 MPa range. These exceptional properties make UHPC increasingly attractive for bridge decks, blast-resistant structures, and infrastructure applications where durability and strength-to-weight ratios are critical.

However, designing UHPC mixes remains a complex undertaking. A typical formulation involves cement, supplementary cementitious materials such as silica fume and fly ash, fine aggregates, water, superplasticizer, and in many cases steel or synthetic fiber reinforcement. Each ingredient interacts with the others in ways that are difficult to predict from first principles alone. The water-to-cement ratio, for example, is one of the most fundamental relationships in concrete science: lower ratios generally produce stronger concrete because less excess water means fewer capillary pores in the hardite matrix. But this relationship interacts with superplasticizer dosage (which allows workability at lower water content), aggregate proportions (which affect the overall paste volume), and supplementary cementitious materials (which participate in secondary hydration reactions). Curing conditions—particularly temperature and age at testing—add further complexity. This is precisely the kind of problem where machine learning can add value: learning multi-variable, non-linear relationships from historical data to predict outcomes for new formulations.

> ***Expansion Opportunity:*** *A literature review section could survey prior ML approaches to concrete strength prediction (e.g., Chou et al. 2011, DeRousseau et al. 2019, Feng et al. 2020). Positioning this work relative to published benchmarks would strengthen the academic framing and demonstrate awareness of the broader research landscape.*

1.2 Project Scope and Objectives

This project developed a predictive model for UHPC compressive strength using a publicly available dataset from Mendeley Data (Kashem et al., 2023) containing 810 mix designs. The primary objective was to identify a regression model whose prediction error falls within an acceptable range—approximately 5–10% of the mean compressive strength—while providing interpretable insights into which mix components most influence strength.

1.3 Pipeline Design Philosophy

The analysis followed a structured three-notebook pipeline, a deliberate architectural decision that merits explanation. Each notebook represents a distinct analytical phase with well-defined inputs and outputs:

Notebook 01 (Exploratory Analysis) ingests the raw CSV, validates data quality, explores distributions and relationships, and exports a cleaned CSV. It makes no modeling decisions—it only observes and documents.

Notebook 02 (Model Development) ingests the cleaned CSV, trains and compares models, performs hyperparameter tuning, and exports the final trained model as a joblib artifact along with explicit train/test split CSVs.

Notebook 03 (Model Interpretation) ingests the saved model and the exact same train/test splits, ensuring that interpretation is performed on identical data without any risk of random seed mismatches.

This separation matters for several reasons. First, it enforces reproducibility: anyone can re-run Notebook 03 against the saved model without retraining, and the results will be identical. Second, it creates a clean audit trail: if a stakeholder questions the model's behavior, the interpretation notebook can be examined independently of the training code. Third, it reflects professional data science practice, where production pipelines are modular rather than monolithic. The use of explicit file handoffs (saved CSVs and joblib artifacts) rather than relying on shared random seeds is a deliberate robustness choice—if a library update changed the random number generator, the pipeline would still produce identical interpretation results.

Constants were defined centrally in each notebook (RANDOM_STATE = 42, UHPC_THRESHOLD = 150, feature and target names) and kept consistent across all three. This is a small but important discipline that prevents the kind of subtle bugs that arise when magic numbers are scattered through code.

> ***Expansion Opportunity:*** *Adding a configuration file (e.g., config.yaml) shared across all three notebooks would further centralize constants and eliminate any possibility of drift. A Makefile or pipeline orchestrator (e.g., DVC, Prefect) could automate the end-to-end flow and ensure notebooks execute in the correct order.*

2. Technical Decisions and Methodology Rationale

Before walking through the analysis results, it is worth pausing to explain the reasoning behind several key technical decisions. In data science, the “why” behind each choice often matters as much as the results themselves, because it reveals whether the analyst understood the tradeoffs involved.

2.1 Why RMSE as the Primary Metric

Three evaluation metrics were tracked throughout the project: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). Each tells a different story.

RMSE was chosen as the primary comparison metric for two reasons. First, it is expressed in the same units as the target variable (MPa), making it directly interpretable—an RMSE of 5.93 MPa means the model’s typical prediction is off by about 6 MPa. Second, RMSE penalizes large errors more heavily than MAE due to the squaring operation. In an engineering context, a prediction that is 30 MPa off is far more consequential than three predictions that are each 10 MPa off (imagine specifying a bridge component based on a 30 MPa overestimate), and RMSE captures this asymmetry in consequence.

MAE was included alongside RMSE because it provides a complementary perspective. MAE treats all errors equally regardless of magnitude, which makes it more robust to outliers and easier to explain to non-technical audiences ("the model is off by about 4 MPa on average"). The ratio between RMSE and MAE also provides diagnostic information: when RMSE is substantially larger than MAE, it signals the presence of a few large errors pulling up the RMSE. In the final model, RMSE (5.93) and MAE (4.25) had a ratio of 1.40, indicating moderately uniform error distribution with some larger misses.

R^2 was tracked for completeness but not used as the primary metric because it can be misleading. An R^2 of 0.978 sounds impressive, but it does not directly tell you the magnitude of errors in engineering units. It is also sensitive to the variance of the target: the same model applied to a dataset with less variation in compressive strength would show a lower R^2 even if the absolute prediction errors were identical. RMSE avoids this issue.

2.2 Why Five-Fold Cross-Validation

Five-fold cross-validation was selected as the evaluation framework, using scikit-learn’s KFold with shuffle=True and random_state=42. This choice involves several layered decisions worth unpacking.

The number of folds (k=5) represents a standard bias-variance tradeoff. Fewer folds (e.g., k=3) train on less data per fold, which increases bias in the performance estimate but reduces computational cost. More folds (e.g., k=10 or leave-one-out) train on more data per fold, reducing bias but increasing variance in the estimate and computational time. With n=633 training samples, five folds provide training sets of approximately 506 samples each—large

enough for stable model fitting while still leaving 127 samples per validation fold for reasonable error estimation.

Shuffling was enabled to prevent any ordering artifacts in the data. Without shuffling, if the original dataset happened to be sorted by strength (or by submission date, which might correlate with strength through lab practices), different folds would contain systematically different subpopulations, biasing the cross-validation estimate.

An important nuance: the cross-validation used regular KFold rather than StratifiedKFold. Stratification was applied at the train/test split level (ensuring both sets had ~26.5% UHPC-grade samples) but not within the cross-validation folds. For regression problems, StratifiedKFold is less commonly used because it requires binning the continuous target, and the shuffling already provides reasonable distributional balance across folds. If cross-validation variance had been a concern, stratified folds could have been tested as a refinement.

> ***Expansion Opportunity:*** *Comparing 5-fold vs. 10-fold vs. repeated 5-fold (e.g., 5×2 = 10 total fits) would quantify how sensitive the results are to the CV scheme. If RMSE rankings remained stable across all schemes, it would strengthen confidence in the model selection. Repeated cross-validation is particularly valuable for small datasets.*

2.3 Why StandardScaler Inside a Pipeline

All linear models (Linear Regression, Ridge, LASSO) were wrapped inside scikit-learn Pipelines that included StandardScaler as a preprocessing step. This is a critical design choice that prevents a subtle but serious form of data leakage.

StandardScaler transforms features to have zero mean and unit variance by computing the mean and standard deviation of each feature. If scaling were applied to the entire training set before cross-validation, the validation folds would be scaled using statistics (means and standard deviations) that were computed from data that includes those very validation samples. This contaminates the validation estimate: the model has indirectly “seen” the validation data through the scaling parameters. The leakage is typically small for scaling, but the principle matters—and in other preprocessing steps like target encoding or imputation, the leakage can be severe.

By placing StandardScaler inside a Pipeline, scikit-learn’s cross_validate function automatically fits the scaler on each training fold and transforms the corresponding validation fold using only training-fold statistics. This mirrors what would happen in production: when a new sample arrives, it would be scaled using the training set’s statistics, not its own.

Tree-based models (Random Forest, XGBoost) were intentionally excluded from this pipeline. Decision trees make splits based on threshold comparisons (e.g., “is cement > 750?”), which are invariant to linear scaling. Scaling would produce identical tree structures and predictions, adding computational overhead for no benefit.

2.4 Why Flag-and-Filter for Data Cleaning

The data quality validation in Notebook 01 used a flag-and-filter approach: each type of quality issue (negatives, out-of-range values, duplicates) was flagged in a separate boolean column, and then all flagged rows were removed simultaneously at the end. The alternative would have been to drop rows as each issue was detected.

The flag-and-filter approach was chosen for transparency and reproducibility. By creating explicit flag columns, the notebook preserved a complete record of which rows were flagged and why. If someone wanted to challenge the cleaning decisions—for example, arguing that duplicates should be retained because they represent valid repeated experiments—they could re-run the notebook with a different filter expression without modifying the validation logic. The helper columns were dropped before saving the cleaned dataset, so they did not propagate into modeling.

2.5 Why These Five Models

The selection of five candidate models—Linear Regression, Ridge, LASSO, Random Forest, and XGBoost—was deliberately structured to test specific hypotheses raised during exploratory analysis.

Linear Regression served as the simplest possible baseline. Its role was not to win the comparison but to establish a floor: if a model cannot outperform ordinary least squares, it is not worth the added complexity.

Ridge (L2 regularization) was included specifically to test whether the severe multicollinearity identified in EDA (VIF > 50 for water and cement) was degrading linear model performance. Ridge shrinks correlated coefficients toward each other, which can stabilize predictions when features are collinear.

LASSO (L1 regularization) tested whether implicit feature selection would help. With 13 features and some showing weak target correlations (fly_ash: r = -0.05, slag: r = 0.04), LASSO's ability to zero out irrelevant coefficients could potentially improve generalization by reducing model complexity.

Random Forest was the first non-linear model. Tree ensembles naturally handle the zero-inflation, skewness, and multicollinearity that characterized this dataset. Random Forest's bagging approach (training many trees on bootstrap samples of the data and averaging their predictions) also provides built-in variance reduction.

XGBoost was included as a more sophisticated tree-based approach. Unlike Random Forest, which trains trees independently, XGBoost builds trees sequentially where each new tree specifically targets the residual errors of the previous trees. This gradient boosting mechanism can be more efficient at capturing complex patterns because each tree focuses on the hardest-to-predict samples rather than the full data distribution.

> ***Expansion Opportunity:*** *Adding a neural network (e.g., a simple feedforward MLP via scikit-learn's MLPRegressor) would round out the model families and test whether deep learning offers any advantage on this dataset size. Elastic Net (combining L1 and L2*

> *penalties) could also replace or supplement the separate Ridge and LASSO evaluations. Support Vector Regression (SVR) with an RBF kernel is another non-linear option popular in materials science literature.*

3. Phase 1: Exploratory Data Analysis

The exploratory analysis served as the foundation for all subsequent modeling work. Its purpose was threefold: to verify the integrity of the data, to understand the distributional characteristics of both the target and the features, and to identify statistical patterns—particularly multicollinearity—that would influence the choice of modeling approach.

3.1 Dataset Overview and Data Quality

The raw dataset consisted of 810 UHPC mix designs, each described by 14 columns: 13 input features and one target variable. The original column names were single-letter or two-letter abbreviations (C, W, FA, SF, etc.), which were immediately renamed to descriptive labels (cement, water, fly_ash, silica_fume, etc.) for readability. The features covered material quantities (cement, slag, silica_fume, limestone_powder, quartz_powder, fly_ash, nano_silica, aggregate, water, fiber, and superplasticizer, all measured in kg/m³), curing temperature (°C), and age at testing (days). The target variable was compressive strength in megapascals (MPa).

Missing Values

A column-by-column check confirmed zero missing values across all 14 columns. This is relatively unusual for research datasets and suggests the data was compiled from complete experimental records rather than merged from multiple sources with differing schemas.

Negative and Out-of-Range Values

Every numeric value was screened for physical plausibility in two passes. First, a sweep for negative values found none—all quantities were non-negative, as expected for physical measurements. Second, each feature was validated against literature-derived ranges. These ranges were intentionally set wider than typical values to avoid flagging legitimate experimental data: for example, the cement range was set at 200–1500 kg/m³ even though typical UHPC formulations use 700–1000 kg/m³. Sources for the validation ranges included FHWA-HRT-13-060, PMC9228538, PMC5455500, and other published literature. No values fell outside these ranges.

> ***Expansion Opportunity:*** *The validation ranges could be tightened to flag "unusual but technically valid" observations that might represent atypical formulations. A more advanced approach would compute Mahalanobis distances or use isolation forests to identify multivariate outliers—rows that are individually within range on every feature but collectively represent unusual combinations. For instance, a mix with very high cement AND very high water might be unusual even if each value alone is plausible.*

Duplicate Records

An exact-match duplicate check across all original columns identified 18 duplicate rows (2.2% of the dataset). These duplicates were flagged but deserve careful consideration. In experimental datasets, duplicates can represent either (a) repeated experiments intentionally conducted to assess reproducibility, or (b) data entry errors or copy-paste mistakes during data compilation. The dataset documentation did not clarify which scenario applied. The conservative decision

was to remove duplicates to prevent the model from seeing identical data points in both training and validation folds, which would artificially inflate performance estimates. The cleaned dataset retained 792 of the original 810 records (97.8% retention rate).

Summary Statistics

Summary statistics revealed wide variation in feature magnitudes. Aggregate values ranged from 408 to 1,992 kg/m³, while nano-silica ranged from 0 to 47.5 kg/m³—a 40-fold difference in scale. This magnitude disparity directly motivated the decision to use StandardScaler for linear models: without scaling, a coefficient-based model would give disproportionate weight to features with larger numeric values simply because of their units.

The summary statistics also quantified zero-inflation by comparing total records to non-zero records for each feature. Cement, aggregate, water, superplasticizer, temperature, and age all had 810 non-zero records—they appear in every mix. In contrast, slag had only 98 non-zero records (12%), fly_ash had 117 (14%), and nano-silica had 211 (26%). These features were absent in the majority of formulations, creating highly skewed distributions dominated by zeros.

3.2 Target Variable Analysis

Distribution Shape

A histogram with kernel density estimation showed compressive strength following an approximately symmetric, unimodal distribution centered near 120 MPa. The range spanned from 28.5 MPa to 220.5 MPa, covering everything from conventional high-strength concrete to top-tier UHPC. The interquartile range (IQR) of 56.4 MPa (Q1 = 94.6, Q3 = 151.0) indicated healthy spread—the dataset was not concentrated at any particular strength level, which is important for the model to learn across the full performance range.

Normality Assessment

Normality was assessed through three complementary lenses. Quantitative shape statistics showed a skewness of 0.034 (essentially zero, indicating perfect symmetry) and a kurtosis of -0.506 (within the near-normal range of $|k| < 1$, indicating tails that are slightly lighter than a true Gaussian). A Q-Q plot showed data points tracking closely along the reference diagonal, with only minor deviations at the extreme tails—behavior that is well within expected range for n = 792.

However, the D'Agostino-Pearson omnibus test formally rejected normality (test statistic = 15.29, p = 0.0005). This apparent contradiction between the visual/quantitative evidence (near-normal) and the formal test (not normal) is a textbook example of a well-known phenomenon: with large sample sizes, formal normality tests become overpowered and reject even trivial departures from perfect normality. At n = 792, the test detected real but practically irrelevant deviations that the shape statistics quantified as negligible.

The practical conclusion was that the target distribution was sufficiently well-behaved for modeling purposes and did not require transformation. A note was made to reassess residual normality during model evaluation to confirm this assumption held.

> ***Expansion Opportunity:*** *Adding a Shapiro-Wilk test (more powerful for moderate samples) or a Kolmogorov-Smirnov test with Lilliefors correction would provide additional normality evidence. More importantly, plotting the distribution of residuals from the final model would directly test the assumption that matters—not whether the target itself is normal, but whether prediction errors are approximately normal.*

UHPC Threshold Analysis

The 150 MPa UHPC threshold divided the dataset into 210 samples (26.5%) at or above the threshold and 582 samples (73.5%) below it. The mean strength above the threshold was 172.8 MPa compared to 103.8 MPa below it. This imbalance—not in a classification sense, but in terms of the density of training examples across the target range—meant the model would have roughly three times more sub-threshold examples to learn from than UHPC-grade examples. This could potentially reduce accuracy at the high end where training examples are sparser.

To mitigate this, the train/test split in Notebook 02 was stratified using the UHPC threshold as a binary indicator. This ensured both the training set (26.5% UHPC-grade) and test set (26.4%

UHPC-grade) maintained representative distributions, preventing a situation where the test set might by chance contain a disproportionately easy or hard mix of strength levels.

> ***Expansion Opportunity:*** *A classification layer could be added alongside the regression model: train a binary classifier to predict whether a mix meets the 150 MPa threshold, then evaluate precision/recall/F1. This would directly answer an engineering question ("will this mix qualify as UHPC?") and would be a valuable complement to the continuous strength prediction.*

3.3 Feature Distributions and Characteristics

Distribution Patterns

A grid of 13 histograms with KDE overlays revealed two distinct distributional families. The required components—cement, water, aggregate, and superplasticizer—appeared in every mix design (zero-value rate of 0%) and exhibited continuous, roughly bell-shaped distributions with varying degrees of skewness. Cement showed a broad distribution centered around 770 kg/m³, water was tightly concentrated between 160–200 kg/m³, aggregate spanned widely from 400–2000 kg/m³, and superplasticizer showed a nearly uniform distribution between 1–57 kg/m³.

The optional components—slag, fly ash, nano-silica, limestone powder, quartz powder, and fiber—showed dramatic zero-inflation. Their histograms were dominated by a large spike at zero (representing mixes that omitted the ingredient), with a secondary distribution among the non-zero values. Zero-value rates ranged from 22% for silica fume up to 92% for fly ash. This zero-inflated pattern has significant modeling implications: standard transformations like the logarithm cannot handle zero values (log(0) is undefined), and even the common log(x+1) workaround can distort the signal, particularly when zeros carry meaningful information (the ingredient was deliberately excluded, not merely present in small amounts).

Outlier Assessment

Boxplots of non-zero values confirmed the scale differences seen in histograms and revealed moderate outliers in temperature (values up to 210°C, representing heat-cured specimens) and age (values up to 365 days, representing long-term curing tests). These were assessed as legitimate experimental conditions rather than data errors: heat curing is a standard practice for accelerating UHPC strength development, and 365-day testing is routine for durability assessment. No outlier removal was performed, preserving the full range of experimental conditions for the model to learn from.

> ***Expansion Opportunity:*** *IQR-based outlier detection could be formalized (e.g., values beyond 1.5×IQR or 3×IQR from Q1/Q3), and a sensitivity analysis could compare model performance with and without extreme values. Cook's distance could be calculated after fitting the linear baseline to identify observations with outsized influence on regression coefficients.*

Skewness Analysis

Absolute skewness was computed for all 13 features. Seven exceeded the |skew| > 1 threshold commonly used to flag non-normal distributions: temperature (9.04), limestone_powder (4.70), age (3.89), slag (3.18), nano_silica (2.49), fly_ash (2.45), and quartz_powder (2.31). Temperature's extreme skewness (9.04) reflected the concentration of most experiments near room temperature (20–23°C) with a few outlying heat-cured specimens. The other highly skewed features were all optional components whose skewness was driven by the zero-inflation pattern.

These features were flagged as potential transformation candidates, but the decision to transform was intentionally deferred to the modeling phase. The reasoning was straightforward: tree-based models (Random Forest, XGBoost) make binary splits on feature values and are therefore invariant to monotonic transformations—transforming a feature changes its numeric values but not the split ordering, so the model learns identical structure regardless. If tree-based models were to outperform linear approaches (as the EDA patterns suggested they would), transformations would be unnecessary.

3.4 Feature-Target Relationships

Correlation Analysis

Pearson correlation coefficients quantified the linear association between each feature and compressive strength. The strongest correlations were moderate in magnitude. Silica fume led at r = 0.53, followed by fiber (r = 0.45), aggregate (r = -0.43), superplasticizer (r = 0.41), age (r = 0.41), and cement (r = 0.34). Six features fell in the moderate correlation range (|r| = 0.3–0.7), while the remaining seven showed weak linear associations.

No feature achieved the |r| > 0.70 threshold often used in engineering contexts to indicate a strong linear relationship. This was a significant finding: it suggested that compressive strength is not strongly linearly dependent on any single ingredient. The relationships are either non-linear (e.g., there may be an optimal cement dosage rather than a purely monotonic one), involve interactions between features (e.g., silica fume's effect may depend on cement content), or both. This provided early evidence favoring non-linear models.

Statistical Significance

Significance testing (Pearson p-values at α = 0.05) found that 11 of 13 features had statistically significant correlations with the target. The two non-significant features were fly ash (r = -0.05, p = 0.17) and slag (r = 0.04, p = 0.29). However, the notebook appropriately cautioned against over-interpreting p-values at this sample size. With n = 792, even very small correlations achieve significance: for example, a correlation of r = 0.07 is statistically significant at α = 0.05 with n = 792, despite explaining less than 0.5% of the variance. Statistical significance is a necessary but insufficient condition for practical importance.

Scatter Plot Assessment

Scatter plots for the top five correlated features plotted against compressive strength showed approximately linear trends with substantial scatter. No obvious heteroscedasticity was observed—the spread of residuals appeared roughly constant across each predictor's range—which was reassuring for linear modeling assumptions. The fiber scatter plot showed a distinctive two-population pattern: a vertical cluster at fiber = 0 (the 54% of mixes without fiber) showing the full range of strength outcomes, followed by a positive trend among the non-zero fiber values. This pattern—where a feature is either absent or present with a continuous positive effect—is a natural fit for tree-based models, which can first split on "fiber = 0 vs. fiber > 0" and then learn the continuous relationship within the non-zero group.

> ***Expansion Opportunity:*** *Spearman rank correlations (which capture monotonic but non-linear relationships) could be computed alongside Pearson correlations. A comparison between the two would directly quantify how much non-linearity exists in each feature-target relationship. Features where Spearman >> Pearson would be the strongest candidates for non-linear modeling. Additionally, partial dependence plots or locally weighted scatterplot smoothing (LOWESS) curves could visualize the actual functional form of each relationship.*

3.5 Multicollinearity Assessment

Pairwise Correlations

A correlation heatmap visualized all pairwise relationships among the 13 features. The strongest inter-feature correlation was between silica fume and aggregate at r = -0.70, right at the boundary of the conventional concern threshold. Other notable pairs included aggregate and water (r = -0.55), cement and water (r = 0.50), cement and aggregate (r = -0.48), and cement and slag (r = -0.46). When all 78 unique feature pairs were tabulated, 14 fell in the moderate range (|r| = 0.3–0.7) and 64 were weak (|r| < 0.3). Crucially, no pairs exceeded |r| = 0.70, which at first glance suggested multicollinearity was not a major concern.

Variance Inflation Factors

VIF analysis revealed a more concerning picture. VIF measures how much the variance of a regression coefficient is inflated due to linear dependencies with all other features simultaneously—not just pairwise relationships. A feature with high VIF can be substantially predicted from a combination of other features, even if no single pairwise correlation is strong.

Using the Hair et al. (2010) thresholds (VIF ≤ 5 acceptable, 5–10 moderate, > 10 severe), three features showed severe multicollinearity: water (VIF = 72.71), cement (VIF = 59.12), and aggregate (VIF = 13.37). Two additional features showed moderate concern: superplasticizer (VIF = 8.64) and silica fume (VIF = 5.88). The remaining eight features had VIF values below 5.

The extremely high VIF values for water and cement—despite neither having a pairwise correlation above 0.55 with any single other feature—revealed that these variables form a multicollinear cluster. When considered jointly, cement, water, and aggregate can predict each other with high accuracy because concrete mix design is governed by ratios: the water-to-cement ratio (w/c) typically falls in a narrow range for UHPC (0.15–0.25), and aggregate proportions are determined relative to the paste volume. These physical constraints create inherent linear dependencies that VIF detects even when no single pairwise correlation is alarming.

The multicollinearity findings had direct implications for model selection. Ordinary least squares regression would produce unstable, hard-to-interpret coefficients for the collinear features—small changes in the data could cause large swings in the cement and water coefficients while their combined effect remained stable. Ridge regularization could mitigate this instability. Tree-based models, however, are fundamentally unaffected because they make sequential splits on individual features rather than estimating simultaneous coefficients. This finding, combined with the zero-inflation and skewness patterns, provided strong a priori justification for tree-based models.

> ***Expansion Opportunity:*** *Creating derived ratio features—particularly water-to-cement ratio (w/c) and water-to-binder ratio (w/b, where binder includes cement plus all supplementary cementitious materials)—would directly encode the domain knowledge that these ratios are more physically meaningful than the raw quantities. This could reduce multicollinearity and potentially improve both linear and tree-based model*

> *performance. Another option: Principal Component Analysis (PCA) applied to the collinear cluster to create orthogonal composite features.*

4. Phase 2: Model Development

The model development phase translated the patterns and concerns identified during exploratory analysis into a structured comparison of five regression approaches. The design philosophy was to start with simple, interpretable baselines and progressively move to more flexible models, evaluating each under the same cross-validation framework to enable fair comparison.

4.1 Data Preparation

The cleaned dataset of 792 records was split into a training set of 633 samples (80%) and a held-out test set of 159 samples (20%). The split was stratified using a binary indicator derived from the 150 MPa UHPC threshold, ensuring the training set contained 26.5% UHPC-grade samples and the test set contained 26.4%—an almost exact match. The test set was then set aside entirely and not touched until the final evaluation, ensuring it provided a genuinely unbiased estimate of out-of-sample performance.

Both the feature matrices and target vectors for train and test sets were saved to CSV files after splitting. This explicit export was a deliberate reproducibility measure: Notebook 03 (interpretation) would load these exact files rather than recreating the split, eliminating any risk that a different library version or platform might produce a different random split even with the same seed.

4.2 Baseline Model Comparison

Linear Model Results

The three linear approaches produced remarkably similar results, which itself told an important story.

Linear Regression achieved a cross-validation RMSE of 22.76 MPa (±1.29), MAE of 17.98 MPa, and R^2 of 0.6684. This baseline established the floor: a simple additive model with no regularization or non-linear terms explained about two-thirds of the variance in compressive strength. The RMSE of 22.76 represents roughly 19% of the mean target value (122 MPa)—functional for rough estimation but not precise enough for engineering specification.

Ridge Regression achieved an essentially identical RMSE of 22.76 MPa (±1.26) and R^2 of 0.6684. This result is informative precisely because Ridge was expected to help. Ridge's L2 penalty shrinks coefficients toward zero, which typically stabilizes predictions when features are collinear. The fact that it made no measurable difference suggests that the multicollinearity identified in EDA, while real, was not the primary bottleneck in prediction accuracy. The collinear features (cement, water, aggregate) collectively carried the same predictive information regardless of how their individual coefficients were distributed. The limitation was the linear functional form itself, not coefficient instability.

LASSO Regression performed slightly worse: RMSE of 23.50 MPa (±0.88), R^2 of 0.6471. LASSO's L1 penalty drives some coefficients exactly to zero, performing implicit feature

selection. The degraded performance indicates that even the weakly correlated features (those with $r < 0.20$) contributed marginal predictive information, and dropping them hurt more than it helped. Notably, LASSO's cross-validation standard deviation was the lowest of any model (0.88 vs. 1.07–1.29 for others), suggesting that its simpler model (with some features zeroed out) was more stable across folds even if less accurate on average.

Taken together, the linear model results provided two key insights. First, the similarity across all three variants indicated that the performance ceiling was set by the linear functional form, not by multicollinearity or irrelevant features. Second, explaining only 67% of the variance left a large unexplained component—roughly one-third of the variation in compressive strength could not be captured by any additive combination of the 13 features. This strongly suggested non-linear relationships and feature interactions.

Tree-Based Model Results

The transition from linear to tree-based models produced the most dramatic result in the entire analysis.

Random Forest achieved an RMSE of 9.30 MPa (±0.85) and R^2 of 0.9437—a 59% reduction in RMSE compared to the linear baseline. Random Forest works by training hundreds of decision trees on bootstrap samples of the training data, with each tree considering a random subset of features at each split. The final prediction is the average of all trees' individual predictions. This bagging approach reduces variance (each individual tree may overfit, but their average does not) while the random feature selection decorrelates the trees and ensures diversity in the ensemble.

XGBoost achieved the best baseline performance with an RMSE of 8.41 MPa (±1.07) and R^2 of 0.9533—a 63% reduction in RMSE over the linear baseline and a 10% improvement over Random Forest. XGBoost's advantage comes from its sequential boosting mechanism. While Random Forest builds trees independently (each tree sees a bootstrap sample but doesn't know what the other trees predicted), XGBoost builds trees one at a time, with each new tree specifically targeting the residual errors from the ensemble so far. The first tree captures the broadest patterns; the second tree focuses on what the first tree got wrong; the third tree corrects the second tree's remaining errors; and so on. This focused error-correction approach typically requires fewer trees to achieve the same accuracy and can capture more subtle patterns.

The performance gap between the two model families was visually striking in the cross-validation comparison bar chart. The three linear models clustered together at RMSE values above 22 MPa, separated by a wide gap from the two tree-based models clustered below 10 MPa. This was not a marginal difference requiring careful statistical testing—it was a qualitative leap in predictive capability, indicating that the non-linear relationships and feature interactions identified during EDA were not minor effects but were responsible for approximately one-third of the total variation in compressive strength.

Table 1: *Cross-Validation Results — All Baseline Models*

Model	RMSE	RMSE σ	MAE	MAE σ	R^2	R^2 σ
Linear Regression	22.76	1.29	17.98	0.96	0.668	0.05
Ridge	22.76	1.26	17.99	0.94	0.668	0.05
LASSO	23.50	0.88	18.92	0.63	0.647	0.05
Random Forest	9.30	0.85	6.78	0.52	0.944	0.01
XGBoost	**8.41**	**1.07**	**6.05**	**0.63**	**0.953**	**0.01**

4.3 Hyperparameter Tuning

Based on the baseline results, XGBoost and Random Forest were selected for hyperparameter tuning using GridSearchCV with the same five-fold cross-validation framework. Linear models were excluded from tuning because their performance ceiling was determined by the linear functional form, not by hyperparameter settings.

XGBoost Tuning

A grid of 27 parameter combinations was evaluated across three key hyperparameters. The number of boosting rounds (n_estimators) was tested at 100, 200, and 300—controlling how many sequential trees the model builds. Maximum tree depth (max_depth) was tested at 3, 5, and 7—controlling how complex each individual tree can be. Learning rate was tested at 0.05, 0.1, and 0.2—controlling how much each tree's contribution is scaled down before adding it to the ensemble. Lower learning rates generally require more trees but produce smoother, more generalizable models.

The optimal configuration was 300 estimators, a maximum depth of 5, and a learning rate of 0.2, achieving a cross-validation RMSE of 7.98 MPa. This configuration makes intuitive sense. A depth of 5 allows each tree to capture interactions involving up to 5 features—sufficient for complex multi-way interactions without creating overly specialized trees that memorize training patterns. The learning rate of 0.2 is at the higher end of the tested range, suggesting the model benefits from more aggressive boosting steps. With 300 trees at a 0.2 learning rate, the effective model complexity is moderate.

> ***Expansion Opportunity:*** *The tuning grid could be expanded with additional hyperparameters: subsample (row subsampling per tree), colsample_bytree (feature subsampling per tree), min_child_weight (minimum sum of instance weight in a child node), and reg_alpha/reg_lambda (L1/L2 regularization on leaf weights). RandomizedSearchCV over a broader parameter space with more iterations, or Bayesian optimization (e.g., Optuna), could more efficiently explore the hyperparameter landscape than exhaustive grid search.*

Random Forest Tuning

A substantially larger grid of 540 combinations was searched across five hyperparameters: number of trees (100–500), maximum depth (unlimited or 10–30), minimum samples to split a node (2, 5, or 10), minimum samples per leaf (1, 2, or 4), and maximum features per split (square root of total, 50%, or all). Despite this extensive search, the tuned Random Forest achieved an RMSE of 8.86 MPa—improved from its 9.30 baseline but still trailing XGBoost at 7.98 MPa. The optimal parameters (500 trees, max_depth=20, min_samples_split=2, min_samples_leaf=1, max_features=0.5) indicated that the model preferred relatively deep trees with minimal leaf constraints and moderate feature randomization.

The persistent performance gap between tuned XGBoost (7.98) and tuned Random Forest (8.86) reinforced the theoretical expectation: for this dataset, the sequential error-correction

mechanism of gradient boosting extracted more signal than the parallel averaging mechanism of bagging.

4.4 Final Model Evaluation

The tuned XGBoost model was evaluated on the held-out test set of 159 samples that had been completely excluded from all training and tuning decisions.

Test Set Performance

Test set results: RMSE of 5.93 MPa, MAE of 4.25 MPa, and R^2 of 0.978. To put these numbers in engineering context: the model's typical prediction error (MAE) is about 4 MPa, which represents roughly 3.5% of the mean compressive strength. For a concrete with a true strength of 150 MPa (the UHPC threshold), the model's prediction would typically fall between 146 and 154 MPa. Whether this accuracy is sufficient depends on the application: for screening candidate mix designs before laboratory testing, this level of precision could substantially reduce the number of expensive physical tests required. For structural specification, additional safety factors would still be necessary.

The test RMSE (5.93) was notably lower than the cross-validation RMSE (7.98). This is not unusual and does not indicate data leakage or overfitting—it reflects natural variation in the difficulty of different data subsets. The particular 159 samples in the test set happened to be somewhat easier to predict than the average cross-validation fold. What matters is that test performance did not degrade relative to cross-validation, confirming that the model generalizes well to unseen data.

Diagnostic Plots

An actual-versus-predicted scatter plot showed predictions tracking closely along the perfect prediction diagonal across the full strength range from about 30 to 220 MPa. The tightest clustering appeared in the 80–180 MPa range where training data was most dense. Above 200 MPa, predictions showed slightly increased scatter, consistent with the fewer training examples at the high end (recall that only 26.5% of samples exceeded 150 MPa, so the 200+ MPa range had even fewer examples).

A residual plot (actual minus predicted versus predicted values) showed residuals randomly scattered around zero with no systematic patterns. This absence of pattern is itself an important finding—it means the model has no obvious blind spots or systematic biases. There were no funnel shapes (which would indicate heteroscedasticity), no curves (which would indicate missed non-linear structure), and no clusters (which would suggest subpopulation effects). Residual statistics confirmed this: mean of 1.74 MPa (a very slight positive bias, meaning the model tends to slightly underpredict), standard deviation of 5.69 MPa, and a range from -12.89 to +21.31 MPa.

The largest individual error was +21.31 MPa (a sample where the actual strength was 21 MPa higher than predicted). Examining whether such outliers share common characteristics—for

example, unusual combinations of ingredients, extreme curing conditions, or values near the boundary of validation ranges—would help identify where the model is least reliable.

> ***Expansion Opportunity:*** *A predicted-vs-residual analysis segmented by UHPC status (above vs. below 150 MPa) would reveal whether the model performs differently across strength ranges. Plotting residuals against each individual feature could identify features where the model systematically over- or under-predicts. A histogram of residuals with a normality overlay would formally assess whether residual normality holds, which would validate the use of RMSE as the primary metric and support prediction interval construction.*

Bootstrap Prediction Intervals

To quantify prediction uncertainty, 200 bootstrap resamples were used to generate 95% prediction intervals. For each resample, the model was retrained on a bootstrap sample of the training data (drawn with replacement) and used to predict the test set. The spread of 200 predictions for each test sample defined the interval.

The mean prediction interval width was 15.46 MPa, the median was 12.94 MPa, and widths ranged from 4.50 to 48.09 MPa. The variation in interval width is itself informative: narrow intervals correspond to samples in well-represented regions of the feature space (where bootstrap resamples produce consistent models), while wide intervals correspond to samples near the edges of the data or in sparsely populated regions (where different resamples can produce meaningfully different models).

Coverage—the percentage of actual test values falling within the predicted 95% intervals—was 81.8%, substantially below the nominal 95%. This undercoverage is a known limitation of bootstrap prediction intervals for tree-based models. The issue is that bootstrap resampling captures model uncertainty (how the model changes with different training data) but not irreducible noise (the inherent randomness in the target that no model can predict). Additionally, bagged and boosted ensembles tend to reduce variance so effectively that bootstrap predictions cluster together, producing intervals that are too narrow.

> ***Expansion Opportunity:*** *Conformal prediction would address the coverage problem directly. Conformal prediction constructs intervals by calibrating prediction residuals on a held-out calibration set, guaranteeing coverage at any desired level without distributional assumptions. Split conformal prediction is straightforward to implement: reserve a portion of the training data as a calibration set, compute residuals on that set, and use the empirical quantile of those residuals to construct intervals for new predictions. This would be a high-impact addition for engineering applications where knowing the uncertainty bounds is as important as the point prediction.*

5. Phase 3: Model Interpretation

The interpretation phase used SHAP (SHapley Additive exPlanations) to open the black box of the XGBoost model and understand not just which features matter, but how they matter—in what direction, by what magnitude, and for which kinds of samples.

5.1 SHAP Methodology

SHAP values are grounded in cooperative game theory, specifically the concept of Shapley values from 1953. The core idea is to treat each feature as a "player" in a cooperative game where the "payout" is the model's prediction. The Shapley value for each feature is its average marginal contribution across all possible orderings of features—essentially answering the question: "How much does this feature change the prediction, on average, across all possible contexts?"

This framework has several desirable properties that make it superior to simpler alternatives like permutation importance or gain-based importance. First, SHAP values are additive: the base value (average prediction) plus the sum of all SHAP values exactly equals the model's prediction for each individual sample. Second, SHAP fairly distributes credit among correlated features rather than arbitrarily assigning all credit to whichever feature the model happens to split on first. Third, features that don't change the prediction receive a SHAP value of zero, ensuring that irrelevant features are not assigned phantom importance.

For tree-based models, the TreeExplainer algorithm computes exact SHAP values in polynomial time (rather than the exponential time required for exact Shapley values in general), making the computation feasible even for large datasets. The base value for this model was 121.69 MPa—closely matching the training set mean of 122 MPa—which represents the model's prediction before any specific feature information is provided.

> ***Expansion Opportunity:*** *Comparing SHAP-based feature importance with XGBoost's native gain-based importance, permutation importance, and EDA Pearson correlations in a single table would provide a comprehensive view of feature relevance from four different perspectives. Features that rank highly across all methods are the most robust predictors; features that rank differently across methods highlight where the analysis methodology matters.*

5.2 Global Feature Effects

Beeswarm Summary Plot

The SHAP beeswarm plot—arguably the single most informative visualization in the entire analysis—displayed every test observation as a dot for every feature, positioned horizontally by SHAP value and colored by actual feature value. The plot revealed both the magnitude and direction of each feature's effects across the full test set.

Age dominated the plot with the widest SHAP spread, ranging approximately from -40 to +40 MPa. The color pattern was clear and intuitive: red dots (high age, long curing) clustered on the

positive side, while blue dots (low age, early testing) clustered on the negative side. The relationship appeared nearly monotonic—more curing time consistently meant higher predicted strength. The magnitude was remarkable: age alone could shift a prediction by up to 40 MPa in either direction from the 122 MPa base value, effectively determining whether a mix is predicted as standard-strength or UHPC-grade.

Silica fume showed the second-widest spread, with SHAP contributions ranging from roughly -10 to +40 MPa. Higher silica fume content (red dots) consistently pushed predictions upward, reflecting its known role as a pozzolanic material that densifies the cement matrix. The spread was notably asymmetric—more range on the positive side than the negative—suggesting that the benefits of high silica fume content are more pronounced than the penalties of low content. The 22% of samples with zero silica fume clustered near zero SHAP, contributing neither positively nor negatively.

Fiber displayed the most distinctive pattern of any feature. When present in high quantities (red dots), it provided substantial positive lift of +10 to +20 MPa. But a large cluster of blue dots at fiber = 0 sat near the zero SHAP line, representing the 54% of samples with no fiber. This effectively binary pattern—absent with no effect, versus present with a strong positive effect—is precisely the kind of structure that tree-based models handle naturally (the first split can separate zero from non-zero) and that linear models struggle with (a linear model must assign a single coefficient to fiber, which either underpredicts the effect for fiber-containing mixes or overpredicts for fiber-free mixes).

Cement and superplasticizer occupied a middle tier of SHAP importance. Cement's SHAP values showed moderate spread, with higher values generally positive but with some complex non-monotonic behavior—very high cement content did not always produce the highest SHAP contributions, suggesting diminishing returns or an optimal range. Superplasticizer showed a similar moderate spread.

Low-impact features: Fly ash, nano-silica, quartz powder, slag, and limestone powder all showed SHAP values tightly clustered around zero, regardless of their actual values. These features had the highest zero-value rates in the dataset (74–92%) and contributed very little to predictions even when present. This does not necessarily mean these materials have no physical effect on concrete strength; rather, the dataset may lack sufficient variation in these features for the model to learn meaningful relationships. If a future dataset included more formulations that systematically varied these ingredients, their importance might increase.

Mean Absolute SHAP Importance

The SHAP bar plot provided a cleaner ranking by mean absolute impact. The top four features—age, fiber, silica fume, and cement—formed a clear leading group. Superplasticizer and aggregate formed a secondary tier. The remaining seven features contributed relatively little on average.

An interesting discrepancy between the beeswarm and bar plots illuminated a nuance about feature importance. Silica fume showed the second-widest SHAP spread in the beeswarm (suggesting it has the second-largest maximum impact) but ranked third in the bar plot (behind

fiber in mean absolute impact). This indicates that silica fume has very high impact in specific samples—particularly those with high silica fume content—but its average impact across all samples (including the 22% with zero silica fume) is diluted. Fiber, despite being absent in 54% of samples, generates a larger average impact because its non-zero contributions are consistently strong.

The SHAP rankings also shed light on the multicollinearity identified during EDA. Cement, water, and aggregate—the three features with severe VIF values—each showed moderate individual SHAP contributions rather than one dominating at the others' expense. SHAP's game-theoretic framework distributes credit among correlated features based on their marginal contributions, naturally spreading importance across a collinear cluster. This is actually a desirable property: it provides a more honest picture of each feature's role than gain-based importance, which tends to arbitrarily concentrate credit in whichever correlated feature the tree happens to split on first.

5.3 Individual Prediction Analysis

While global analysis reveals overall patterns, waterfall plots for individual predictions show exactly how the model arrives at specific outputs—making the model's reasoning transparent for each sample.

High-Strength Predictions: Two Pathways to UHPC

Two UHPC-grade samples were selected for detailed examination: one without fiber (Sample A, predicted 173 MPa, actual 168 MPa) and one with fiber (Sample B, predicted 214 MPa, actual 201 MPa). The comparison revealed that the model has learned two distinct pathways to high strength.

Sample A (age-driven pathway): This mix was cured for 365 days—a full year—with moderate silica fume (90 kg/m³) and no fiber. The waterfall plot showed age contributing the single largest positive SHAP value, pushing the prediction well above the 122 MPa base. Silica fume added a moderate positive contribution. All other features made relatively small adjustments. The prediction of 173 MPa was achieved almost entirely through extended curing time, demonstrating that patience (allowing hydration reactions to proceed for months) can substitute for aggressive material optimization.

Sample B (composition-driven pathway): This mix was cured for only 28 days—standard testing protocol—but contained very high silica fume (434 kg/m³, near the dataset maximum) and fiber reinforcement at 156 kg/m³. The waterfall plot showed silica fume contributing the largest positive SHAP value, followed by fiber at approximately +17 MPa. Age contributed moderately. The prediction of 214 MPa—the highest in the examined samples—was achieved through optimized material composition rather than extended curing.

This comparison illustrates exactly the kind of multi-pathway structure that tree-based models excel at capturing. A linear model would be forced to assign a single coefficient to age and a single coefficient to silica fume, creating a one-dimensional trade-off. The XGBoost model, through its tree structure, can learn that high age OR high silica fume (or both) lead to high predictions, with the relative importance depending on the context provided by other features.

Low-Strength Predictions: Understanding Weakness

Two low-strength samples were examined to understand what drives weak formulations: the lowest predicted sample (35 MPa predicted, 37 MPa actual, age = 3 days) and a moderate-low sample (63 MPa predicted, 53 MPa actual, age = 1 day).

In both cases, early testing age was the dominant factor pushing predictions downward. The waterfall plots showed age contributing the largest negative SHAP values—mirror images of the positive contributions seen in the high-strength samples. Fiber was absent in both, removing a potential positive contributor. In the lowest-strength sample, cement content also contributed negatively, suggesting a below-optimal dosage.

The moderate-low sample (actual 53 MPa, predicted 63 MPa) is notable because it represents the model's largest prediction error among the examined samples: a 10 MPa overprediction.

The waterfall plot showed that the model expected this 1-day sample to perform better than it did, potentially because other features in the mix (cement, water, aggregate proportions) resembled higher-performing formulations. This kind of error—where the model overestimates the strength of a very young specimen—could occur because the relationship between age and strength at extremely early ages (1–3 days) may not be well represented in the training data, which has a median age of 28 days.

> ***Expansion Opportunity:*** *SHAP interaction values could decompose the contribution of each feature pair, revealing which interactions the model learned. For example, does the effect of silica fume depend on cement content? Does fiber's contribution change at different ages? SHAP dependence plots (SHAP value for one feature plotted against its actual value, colored by a second feature) would visualize these interactions. Additionally, a SHAP force plot covering all 159 test observations would enable interactive exploration, allowing a materials scientist to identify unusual or interesting predictions across the full dataset rather than the four individual samples examined here.*

6. Practical Implications

6.1 Who Would Use This Model

A predictive model for UHPC compressive strength has several potential user communities. Materials researchers could use it to screen candidate mix designs before committing to expensive laboratory casting and testing, which involves procuring raw materials, mixing specimens, curing them for the specified duration, and performing destructive compression testing. A model that can narrow the design space from dozens of candidates to a handful of promising formulations could save significant time and resources.

Quality control engineers at precast concrete facilities could use the model to flag batches whose mix proportions deviate from expected strength targets, triggering additional testing before the concrete is used in structural applications. Design engineers could use the SHAP interpretations to understand which mix modifications would most efficiently increase (or decrease) predicted strength for a specific application.

6.2 What 5.93 MPa Error Means in Practice

The model's RMSE of 5.93 MPa corresponds to approximately 4.9% of the mean compressive strength. In concrete testing, inherent variability is substantial: the coefficient of variation for replicate tests of the same mix design is typically 5–10% for conventional concrete, and UHPC's controlled production can reduce this to 3–5%. This means the model's prediction error is comparable to the inherent testing variability itself. A prediction of 150 ± 6 MPa is within the range of what repeated physical tests of the same mix might produce.

For screening purposes—identifying which candidate formulations are likely to meet the 150 MPa threshold—this accuracy is highly useful. For structural specification, however, additional safety factors and physical testing would still be required. No predictive model should replace actual testing for load-bearing applications.

6.3 Actionable Insights from the Analysis

The analysis produced several findings that translate directly into practical guidance. The dominance of age as a predictor (SHAP range of ±40 MPa) underscores that specifying an appropriate curing duration is the single most impactful design lever. The strong positive effect of silica fume suggests that optimizing its dosage—particularly at levels above 200 kg/m³—could be a high-impact strategy for meeting UHPC thresholds. The binary nature of fiber's contribution (substantial positive effect when present, neutral when absent) suggests that including fiber reinforcement in the formulation may be one of the clearest pathways to improved strength, though the dataset does not distinguish between fiber types.

> ***Expansion Opportunity:*** *An optimization module could be built on top of the model to solve the inverse problem: given a target strength (e.g., 160 MPa), what mix proportions minimize material cost while meeting the target? This could use scipy.optimize with the trained model as the objective function, subject to physical constraints (non-negative*

> *quantities, w/c ratio bounds, total volume constraints). This would transform the project from a predictive tool into a prescriptive one.*

7. Conclusions, Limitations, and Future Work

7.1 Summary of Key Findings

This project demonstrated that gradient-boosted decision trees can predict UHPC compressive strength from mix design parameters with high accuracy. The final XGBoost model achieved an RMSE of 5.93 MPa (approximately 4.9% of mean compressive strength) and R^2 of 0.978 on held-out test data, explaining nearly 98% of the variance in strength outcomes.

The threefold performance gap between linear and tree-based models (RMSE of ~23 vs. ~8 MPa) demonstrated that the relationships between mix ingredients and strength are fundamentally non-linear. Features like fiber and silica fume interact with other components in ways that additive linear terms cannot capture. The EDA patterns that predicted this outcome—zero-inflation, skewness, and multicollinearity—all favored tree-based approaches, and this prediction was borne out.

SHAP analysis revealed that curing age is the single most powerful predictor, capable of shifting predictions by over 40 MPa in either direction. Silica fume and fiber reinforcement round out the top three, with fiber showing a distinctive binary effect pattern. Multiple pathways to UHPC-grade strength exist: extended curing time alone can compensate for moderate material composition, while optimized silica fume and fiber content can achieve even higher strengths at standard testing ages.

7.2 Limitations

Dataset size and scope. At 792 records, the dataset is adequate for initial model development but small by machine learning standards. The limited size constrained the hyperparameter search space and may limit the model's ability to learn subtle interaction effects. More importantly, the dataset represents a specific collection of published UHPC formulations and may not capture the full diversity of mix designs used in practice.

Unspecified material types. The fiber variable does not distinguish between steel fiber, synthetic fiber, glass fiber, or other types—each of which would have substantially different effects on compressive strength. Similarly, the aggregate variable does not specify whether sand, quartz, or other aggregates were used. A model trained on mixed fiber types may produce less accurate predictions for any specific fiber type than a type-specific model would.

Uncertainty quantification. Bootstrap prediction intervals achieved only 81.8% coverage against a 95% target, meaning the model's uncertainty estimates are not currently reliable for decision-making. This is a methodological limitation of bootstrap intervals for tree-based models, not a fundamental limitation of the model itself.

Single random seed. All results depend on a single random seed (42) for data splitting and model initialization. While the stratified split and cross-validation reduce the sensitivity to this choice, a complete robustness analysis would repeat the entire pipeline across multiple seeds.

Interpretation scope. All interpretations are based on model behavior rather than professional materials science expertise. The model's learned patterns should be validated against known cement chemistry and physical mechanisms by domain experts before being used to guide formulation decisions.

7.3 Future Work

Several extensions could strengthen and expand this analysis:

Conformal prediction for properly calibrated uncertainty bounds with guaranteed coverage.

External validation on independent UHPC datasets to test generalizability beyond the training distribution.

Feature engineering with domain-derived features like water-to-cement ratio, water-to-binder ratio, and paste-to-aggregate ratio.

Material type encoding if datasets with specified fiber and aggregate types become available.

SHAP interaction analysis and dependence plots for deeper understanding of multi-feature effects.

Neural network comparison to determine whether deep learning offers advantages at this dataset size.

Mix optimization module that uses the trained model to recommend formulations meeting a target strength at minimum cost.

Multi-seed robustness analysis repeating the pipeline across 10+ random seeds to quantify sensitivity to the data split.

Interactive dashboard (e.g., Streamlit or Tableau) allowing users to input mix proportions and receive predicted strength with uncertainty bounds.

References

Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Lawrence Erlbaum Associates.

Federal Highway Administration. (2013). Ultra-High Performance Concrete: A State-of-the-Art Report for the Bridge Community (FHWA-HRT-13-060).

Federal Highway Administration. (2018). Properties and Behavior of UHPC-Class Materials (FHWA-HRT-18-036).

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2010). Multivariate Data Analysis (7th ed.). Pearson.

Kashem, A., Karim, R., Malo, S., & Hasan, M. (2023). Ultra-High-Performance Concrete (UHPC). Mendeley Data.

Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems, 30.

Lundberg, S. M., et al. (2020). From Local Explanations to Global Understanding with Explainable AI for Trees. Nature Machine Intelligence, 2(1), 56–67.